

TADD MAO

Mathematics & Computer Science, University of Toronto

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RESEARCH INTERESTS

AI verification: how can we know, with formal certainty that an AI system did exactly what we asked? My work approaches this through faithful formalization and formal verification of machine reasoning — building checkers we can trust, rather than trusting the model directly.

EDUCATION

University of Toronto

Sep. 2023 – Apr. 2027 (expected)

H.B.Sc., Computer Science Major (Focus in Artificial Intelligence) & Mathematics Major

Toronto, ON

- Cumulative GPA: **3.99 / 4.00**.
- Selected grades: Mathematical Logic (**100%**), Theory of Computation (99%), Abstract Mathematics (97%), Group Theory (97%), Real Analysis (94%), Advanced Calculus (94%), Machine Learning (90%).
- Coursework: Mathematical Logic, Machine Learning, Artificial Intelligence, Knowledge Representation & Reasoning, Computational Linguistics, Real Analysis, Groups & Symmetries, Topology, Theory of Computation, Programming Languages.

PUBLICATIONS

Does the Proof Prove It That Way? Faithful Formalization of Euclid's Elements Proofs

Tadd Mao, Tianjun Zhong, Dhruva Arekar, Yuming Feng, One An, Jiani Huang, Xujie Si, Ziyang Li.

Preprint; under review at AAAI-27. [arXiv:2608.15432](https://arxiv.org/abs/2608.15432)

- Defined five necessary conditions for how well a formal Lean proof reflects the natural-language proof it claims to represent (faithfulness)
- Built a method that generates proofs meeting these conditions, producing formal Lean proofs more faithful than prior state of the art.

RESEARCH EXPERIENCE

Faithfulness of autoformalized mathematics

June 2026 – present

Johns Hopkins (Ziyang Li) & U. Pennsylvania; with Xujie Si (Toronto)

Collaboration

- First-authored the AAAI-27 submission above (Does the Proof prove it That Way?...)
- Ongoing: a theory-level *faithfulness metric* for AI-generated formalizations of natural-language mathematics (in preparation, targeting ICLR 2027).

Autoformalization in Euclidean Geometry (NSERC USRA)

June 2026 – present

University of Toronto, advised by Prof. Xujie Si

Toronto, ON

- Building the first complete formalization of Euclid's *Elements* in Lean, extending Avigad et al.'s SYSTEM E with a new axiom set.
- Developing methods for evaluating the quality of a given set of axioms.
- Early work on formalizing common-sense reasoning.

Multi-agent reinforcement learning

Aug. 2025 – Apr. 2026

University of Toronto, advised by Prof. Peter Marbach

Toronto, ON

- Studying when *centralized* (one shared policy) versus *decentralized* (per-agent policies) control is preferable in multi-agent RL, characterized by reward variance, task load per agent, and communication under a finite budget.

Laser Scanning Research Assistant

University of Northern British Columbia, under Dr. Wenbo Zheng

Summer 2022

Prince George, BC

- Built the lab's first laser-scanning accuracy metric (custom MATLAB); authored a **workflow guide** that accelerated new-assistant onboarding.

TEACHING

Teaching Assistant — MAT309 (Mathematical Logic)

University of Toronto

Fall 2026

Toronto, ON

- Teaching, marking, and invigilation for the undergraduate mathematical logic course.

HONORS & AWARDS

- NSERC Undergraduate Student Research Award (2026).
- Dean's List Scholar (2024, 2025, 2026 — all completed years).
- Innis College Exceptional Achievement Award (2024).
- Later Life Learning Scholarship (2025).

SELECTED PROJECTS

- **Mathlib contribution (Lean 4)** — authored and merged a lemma (`Int.closure_eq_zmultiples`) into MATHLIB, the community's core formal-mathematics library; wrote the proof independently and refined it through community review and design discussion. [PR #39435](#)
- **Dr. Mario in MIPS assembly** — reimplemented the game from scratch (gravity, collision, pattern-matching, custom audio, difficulty scaling). github.com/tdM05/DrMario
- **Full Stack Chef** — recipe/grocery app (5-person team); led integration and code quality (SOLID, design patterns). github.com/tdM05/FullStackChef
- **Tadd's Museum** — interactive 3D portfolio museum in Unreal Engine 5, with assets sculpted and textured in Blender. [demo](#)

SKILLS

Formal methods: Lean 4, Mathlib

Programming: Python, C/C++

Languages: English, French, Mandarin

